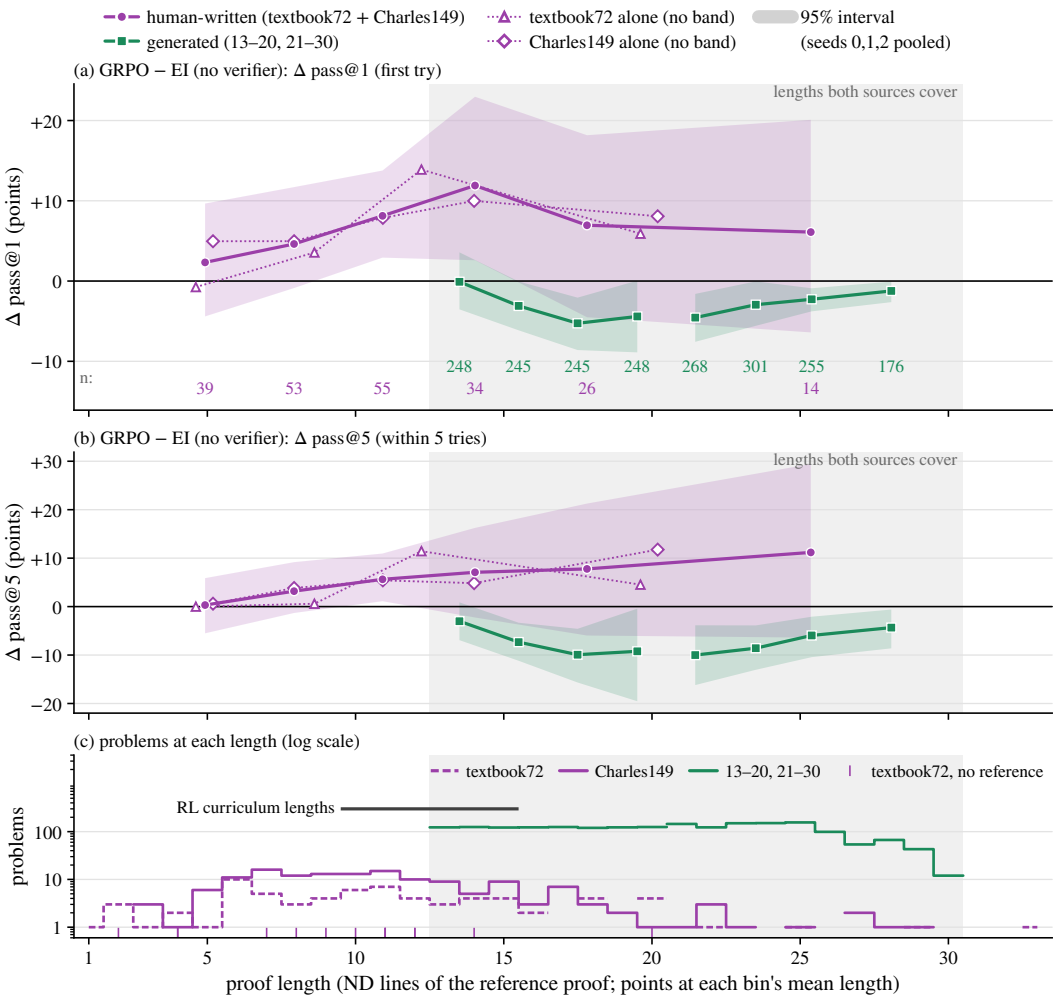
human-written (textbook72 + Charles149)
generated (13–20, 21–30)
textbook72 alone (no band)
Charles149 alone (no band)
95% interval
(seeds 0,1,2 pooled)
(a) GRPO − EI (no verifier): Δ pass@1 (first try)
lengths both sources cover
Δ pass@1 (points)
+20
+10
0
−10
n:
39
53
55
248
34
245
245
26
248
268
301
255
14
176
(b) GRPO − EI (no verifier): Δ pass@5 (within 5 tries)
lengths both sources cover
Δ pass@5 (points)
+30
+20
+10
0
−10
−20
(c) problems at each length (log scale)
textbook72
Charles149
13–20, 21–30
textbook72, no reference
RL curriculum lengths
problems
100
10
1
1
5
10
15
20
25
30
proof length (ND lines of the reference proof; points at each bin's mean length)